MIL-E-1/152C
25 January 1972
SUPERSEDING
MIL-E-1/152B
15 February 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5894

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin tetrode, push-pull rf beam amplifier, F1 = 250 MHz, F2 = 500 MHz

Outline Envelope	}	See figure 1
Base	---	See figure 1 and Group 6 (EIA)
Cathode	---	Unipotential, oxide coated

Base connections:

Pin No.	---	1	2	3	4	5	6	7	Top	leads
Element	---	h	2g1	g2	k int sd	hct	1g1	h	1a	2a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Ik	Ib	Ic1
Unit:	V	Vdc	Vdc	Vdc	v	mAdc	mAdc	mAdc
Maximum:								
C Telep:	6.9/13.8	450	-175	250	100	---	2x80	2x5
C Teleg:	6.9/13.8	600	-175	250	100	2x120	2x110	2x5
Minimum:								
C Telep:	5.7/11.4	---	---	---	---	---	---	---
C Teleg:	5.7/11.4	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3/12.6	600	adjust	250	---	---	40	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Pg1	Pg2	Pp	Pi	TE	Alt
Unit:	W	W	W	W	°C	ft
Maximum:						
C Telep:	2x1	Note 1	2x13.5	2x36	Note 2	10,000
C Teleg:	2x1	7	2x20	2x60	Note 2	10,000
Minimum:						
C Telep:	---	---	---	---	---	---
C Teleg:	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) t = 72 hours

Ⓒ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
(C) 1031	High-frequency vibration	-	t = 900; no voltages	---	---	---	---	---	---
	Quality conformance inspection, part 1								
1301	Heater current	-	Ef = 6.3 V; t = 300	0.65	II	If	1.6	2.0	A
1256	Electrode current (screen grid)	3		0.65	II	Ic2	---	6	mAdc
1261	Electrode voltage (control grid)	3		0.65	II	Ec1	-19	-27	Vdc
1266	Total grid current	3,4	t = 180	0.65	II	Ic1	---	6	μAdc
1231	Peak emission	5	eb = ec1 = ec2 = 225 v	0.65	II	is	1.8	---	a
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	3	Eb = 250 Vdc; Ec1/Ib = 10 mAdc; Rp = 2,000 ohms	---	---	Ep	---	800	mVac
(C) ---	Amplification factor	3,6		---	---	Mu (g2)	6.7	9.6	---
1236	Power oscillation (1)	7,8	Class C amplifier; F = 250 MHz; Eb = 600 Vdc; Ib = 200 mAdc (max); Ic1 = 5 mAdc; Ec2 = 250 Vdc (max); Ic2 = 25 mAdc (max); Rg = 10,000 to 20,000 ohms	---	---	Po	75	---	W
1236	Power oscillation (2)	7,8	Class C amplifier; F = 460 to 490 MHz; Eb = 500 Vdc; Ib = 200 mAdc (max); Ic1 = 6 mAdc (max); Ec2 = 250 Vdc (max); Ic2 = 25 mAdc (max); Rg = 10,000 to 25,000 ohms	---	---	Po	50	---	W
1331	Direct-interelectrode capacitance	9		6.5	Code E	Cgp Cin Cout	0.05 9.40 2.60	0.08 11.60 3.70	pF pF pF
1105	Permanence of marking	-		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
---	Life test	-	Group C; power oscillation (1)	---	---	---	---	---	---
---	Life-test end points (500 hours):								
1266	Total grid current	-		---	---	Ic1	---	-100	μA
1236	Power oscillation (1)	-		---	---	Po	70	---	W
1231	Peak emission	-		---	---	is	1.4	---	a

NOTES:

1. For carrier condition with anode and screen modulation, screen-grid dissipation rating is 4.5 watts. With anode modulation only, screen-grid dissipation rating is 7 watts.
2. Maximum temperature of anode seal = 200°C.
   Maximum temperature of bottom seal = 180°C.
3. Read each unit separately. Control grid of unit not under test shall be connected -100 volts.
4. This test to be performed at the conclusion of the holding period.
5. Both units connected in parallel.
6. Ⓒ Screen grid Mu is determined by measuring grid voltage required to adjust anode current for grid 2 voltages of 250 and 200 volts. (See figure 2.)

   $$Mu = 50/\Delta Ec1 \qquad \text{at } Eb = 600 \text{ Vdc}, \; Ib = 40 \text{ mAdc}$$

7. Power output is total power at drive frequency delivered to load. Drive power is not subtracted from total output.
8. Output circuit adjusted so that anode dissipation does not exceed 2x20 watts.
9. Duplicate test on each unit separately; tie unit not under test to ground.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

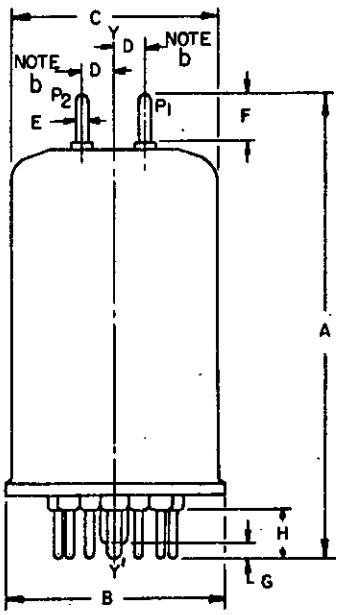
User activities:
Army - WC, MI
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2595)

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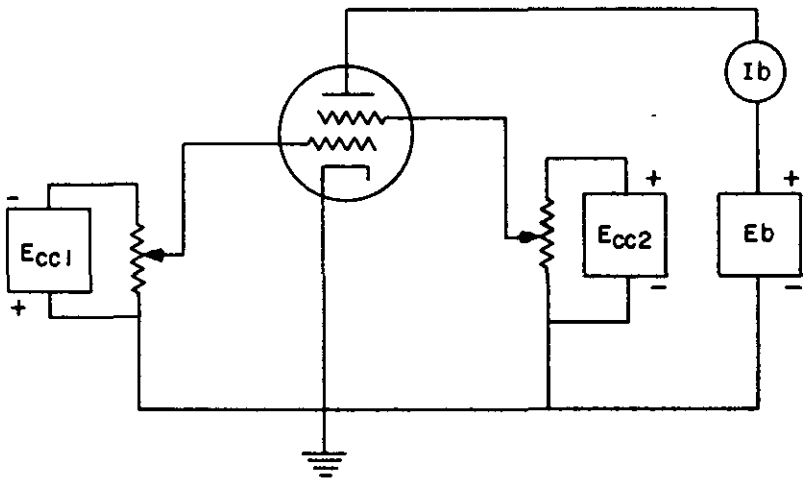


LTR	Dimensions in inches with metric equivalents (mm) in parenthesis	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	3.93 (100.01)	4.31 (109.54)
C	---	1.81 (46.04)
Quality conformance inspection, part 2		
B	---	1.94 (49.21)
D	.275 (6.99) nominal	
E	.077 (1.96)	.083 (2.11)
F	.475 (12.07)	---
G	.000 (00.00)	---
H	.375 (9.53)	.500 (12.70)

NOTES:
a. The axis Y—Y' is defined as the axis of the base pin gage (EIA Group No. 6).
b. The anode leads shall be capable of entering a flat plate gage of .375 ±.001 (9.53 ±.03 mm) thickness having two holes .1400 ±.0005 inch (3.56 ±.01 mm) inch diameter spaced .275 ±.001 inch (6.99 ±.03 mm) from a point coincident with the axis Y—Y'. The axis of the holes shall be parallel to Y—Y' and the plane of these axis shall be 90° ±5' from the plane through Y—Y' and pin No. 4.

© FIGURE 1. Outline drawing of electron tube type 5894.

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© FIGURE 2. Screen-grid amplification factor circuit.

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